

JESS Thermodynamic Database v8.9

Chemical species in reactions with Rhodium

24-Jan-23

Charge	CAS	Count	Molecular formula	Mol. mass
1MeImidazole 1-Methylimidazole; N-Methyl-1,3-diazole; N-Methylimidazole				
0	616-47-7	30	C(4)H(6)N(2)	82.1050
2SHBenz-2				
-2	---	32	C(7)H(4)O(2)S(1)	152.168
Ag+1 Silver(I) ion				
1	14701-21-4	1534	Ag(1)	107.870
Ala-1 Alanine ion; L-Alanine ion; L-2-Aminopropanoate ion				
-1	---	802	C(3)H(6)N(1)O(2)	88.0861
Asn-1 Asparaginate ion; L-2-Aminobutanedioate 4-amide ion; L-beta-Asparaginate ion; Aspartate beta-amide ion; Altheinate ion; Asparamide ion; Agedoite ion				
-1	---	581	C(4)H(7)N(2)O(3)	131.111
bAla-1 beta-Alanate ion				
-1	---	498	C(3)H(6)N(1)O(2)	88.0861
Br-1 Bromide ion				
-1	24959-67-9	1213	Br(1)	79.9040
Cl-1 Chloride ion				
-1	16887-00-6	2242	Cl(1)	35.4530
Cl2(g) Chlorine gas				

0	7782-50-5	589	C1(2)	70.9060
Cys-2 Cysteinate ion; L-Cysteinate ion; beta-mercaptoalanate ion; 2-amino-3-mercaptopropanoate ion; 2-amino-3-mercaptopropionate ion; alpha-amino-beta-thiolpropionate ion				
-2	104170-20-9	673	C(3)H(5)N(1)O(2)S(1)	119.138
e-1 Electron				
-1	---	3399	E(1)	0.000000
Gly-1 Glycinate ion; Aminoacetate ion				
-1	---	1356	C(2)H(4)N(1)O(2)	74.0593
H+1 Hydrogen ion; Proton				
1	12408-02-5	31679	H(1)	1.00794
H+1_PAR-2				
-1	---	46	C(11)H(8)N(3)O(2)	214.203
H+1_TAR-2				
-1	---	23	C(9)H(6)N(3)O(2)S(1)	220.226
H2(g) Hydrogen gas; Hydrogen				
0	1333-74-0	1564	H(2)	2.01588
H2O Water				
0	7732-18-5	5947	H(2)O(1)	18.0153
I-1 Iodide ion				
-1	20461-54-5	894	I(1)	126.900
Ile-1 Isoleucinate ion; L-Isoleucinate ion; 2-amino-3-methylvalerate ion; alpha-amino-beta-methylvalerate ion; 2-amino-3-methylpentanoate ion; ILeu; L-2-amino-3-methylpentanoate ion				

-1	57031-97-7	428	C(6)H(12)N(1)O(2)	130.167
Leu-1 Leucinate ion; L-Leucinate ion; 2-amino-4-methylvalerate ion; alpha-aminoisocaproate ion; 2-amino-4-methylpentanoate ion; L-2-amino-4-methylpentanoate ion				
-1	17332-93-3	511	C(6)H(12)N(1)O(2)	130.167
MeCN Acetonitrile; Cyanomethane; Methyl cyanide; Ethanenitrile				
0	75-05-8	5	C(2)H(3)N(1)	41.0525
Met-1 Methioninate ion; L-Methioninate ion; 2-amino-4-(methylthio)butyrate ion; alpha-amino-gamma-methylmercaptobutyrate ion; 2-amino-4-methylthiobutanoate ion; L-2-amino-4-(methylthio)butanoate ion; gamma-methylthio-alpha-aminobutyrate ion; Meonine ion; Methilalanin ion; Neston ion				
-1	93375-49-6	527	C(5)H(10)N(1)O(2)S(1)	148.200
NH3 Ammonia, aqueous				
0	---	1462	H(3)N(1)	17.0305
O2 (g) Oxygen gas; Oxygen				
0	7782-44-7	2559	O(2)	31.9988
OH-1 Hydroxide ion				
-1	14280-30-9	6558	H(1)O(1)	17.0073
Piperid Piperidine; Pentamethyleneimine; Hexahydropyridine; pip; Perhydropyridine; Azacyclohexane				
0	110-89-4	25	C(5)H(11)N(1)	85.1490
Pyridine Pyridine; Azine; py				
0	110-86-1	297	C(5)H(5)N(1)	79.1014
Rh+1 Rhodium(I) ion				
1	20561-59-5	1	Rh(1)	102.910

Rh+2 Rhodium(II) ion				
2	22541-60-2	10	Rh(1)	102.910
Rh+2_CN-1(6)				
-4	---	1	C(6)N(6)Rh(1)	259.016
Rh+2(2)_1MeImidazole_Butanoic-1(4)				
0	---	2	C(20)H(34)N(2)O(8)Rh(2)	636.319
Rh+2(2)_1MeImidazole(2)_Butanoic-1(4)				
0	---	1	C(24)H(40)N(4)O(8)Rh(2)	718.424
Rh+2(2)_Butanoic-1(4)				
0	---	4	C(16)H(28)O(8)Rh(2)	554.214
Rh+2(2)_MeCN_Butanoic-1(4)				
0	---	2	C(18)H(31)N(1)O(8)Rh(2)	595.266
Rh+2(2)_MeCN(2)_Butanoic-1(4)				
0	---	1	C(20)H(34)N(2)O(8)Rh(2)	636.319
Rh+2(2)_Piperid_Butanoic-1(4)				
0	---	2	C(21)H(39)N(1)O(8)Rh(2)	639.363
Rh+2(2)_Piperid(2)_Butanoic-1(4)				
0	---	1	C(26)H(50)N(2)O(8)Rh(2)	724.512
Rh+2(2)_Pyridine_Butanoic-1(4)				
0	---	2	C(21)H(33)N(1)O(8)Rh(2)	633.315
Rh+2(2)_Pyridine(2)_Butanoic-1(4)				
0	---	1	C(26)H(38)N(2)O(8)Rh(2)	712.416
Rh+3 Rhodium(III) ion				
3	16065-89-7	81	Rh(1)	102.910

Rh+3_2SHBenz-2				
1	---	2	C(7)H(4)O(2)Rh(1)S(1)	255.078
Rh+3_2SHBenz-2(2)				
-1	---	1	C(14)H(8)O(4)Rh(1)S(2)	407.245
Rh+3_Ag+1_NH3(5)_SCN-1				
3	---	1	C(1)H(15)Ag(1)N(6)Rh(1)S(1)	354.010
Rh+3_Ala-1				
2	---	1	C(3)H(6)N(1)O(2)Rh(1)	190.996
Rh+3_Ala-1(2)				
1	---	1	C(6)H(12)N(2)O(4)Rh(1)	279.082
Rh+3_Asn-1				
2	---	1	C(4)H(7)N(2)O(3)Rh(1)	234.021
Rh+3_Asn-1(2)				
1	---	1	C(8)H(14)N(4)O(6)Rh(1)	365.132
Rh+3_bAla-1				
2	---	1	C(3)H(6)N(1)O(2)Rh(1)	190.996
Rh+3_bAla-1(2)				
1	---	1	C(6)H(12)N(2)O(4)Rh(1)	279.082
Rh+3_Br-1				
2	---	1	Br(1)Rh(1)	182.814
Rh+3_Br-1_Cl-1(5)				
-3	---	1	Br(1)Cl(5)Rh(1)	360.079
Rh+3_Br-1(2)				
1	---	1	Br(2)Rh(1)	262.718

Rh+3_Br-1 (3)				
0	---	1	Br (3) Rh (1)	342.622
Rh+3_Br-1 (4)				
-1	---	1	Br (4) Rh (1)	422.526
Rh+3_Br-1 (5)				
-2	---	1	Br (5) Rh (1)	502.430
Rh+3_Cl-1				
2	---	3	Cl (1) Rh (1)	138.363
Rh+3_Cl-1_OH-1				
1	---	1	Cl (1) H (1) O (1) Rh (1)	155.370
Rh+3_Cl-1 (2)				
1	---	3	Cl (2) Rh (1)	173.816
Rh+3_Cl-1 (3)				
0	---	5	Cl (3) Rh (1)	209.269
Rh+3_Cl-1 (3)_(s) Rhodium trichloride				
0	---	1	Cl (3) Rh (1)	209.269
Rh+3_Cl-1 (3)_OH-1				
-1	---	1	Cl (3) H (1) O (1) Rh (1)	226.276
Rh+3_Cl-1 (4)				
-1	---	4	Cl (4) Rh (1)	244.722
Rh+3_Cl-1 (4)_OH-1				
-2	---	1	Cl (4) H (1) O (1) Rh (1)	261.729
Rh+3_Cl-1 (5)				
-2	---	7	Cl (5) Rh (1)	280.175

Rh+3_Cl-1(5)_I-1				
-3	---	1	Cl(5)I(1)Rh(1)	407.075
Rh+3_Cl-1(5)_OH-1				
-3	---	1	Cl(5)H(1)O(1)Rh(1)	297.182
Rh+3_Cl-1(5)_SCN-1				
-3	---	1	C(1)Cl(5)N(1)Rh(1)S(1)	338.253
Rh+3_Cl-1(6)				
-3	---	4	Cl(6)Rh(1)	315.628
Rh+3_CN-1(6)				
-3	---	1	C(6)N(6)Rh(1)	259.016
Rh+3_Cys-2				
1	---	1	C(3)H(5)N(1)O(2)Rh(1)S(1)	222.048
Rh+3_Cys-2(2)				
-1	---	1	C(6)H(10)N(2)O(4)Rh(1)S(2)	341.186
Rh+3_Cys-2(3)				
-3	---	1	C(9)H(15)N(3)O(6)Rh(1)S(3)	460.325
Rh+3_EtDiAm(2)_H2O(2)_(trans)				
3	---	1	C(4)H(20)N(4)O(2)Rh(1)	259.138
Rh+3_EtDiAm(2)_OH-1_H2O_(trans)				
2	---	2	C(4)H(19)N(4)O(2)Rh(1)	258.131
Rh+3_EtDiAm(2)_OH-1(2)_(trans)				
1	---	1	C(4)H(18)N(4)O(2)Rh(1)	257.123
Rh+3_EtDiAm(3)				
3	---	1	C(6)H(24)N(6)Rh(1)	283.207

Rh+3_EtDiAm(3)_SO4-2				
1	---	2	C(6)H(24)N(6)O(4)Rh(1)S(1)	379.264
Rh+3_EtDiAm(3)_SO4-2(2)				
-1	---	1	C(6)H(24)N(6)O(8)Rh(1)S(2)	475.322
Rh+3_Gly-1				
2	---	1	C(2)H(4)N(1)O(2)Rh(1)	176.969
Rh+3_Gly-1(2)				
1	---	1	C(4)H(8)N(2)O(4)Rh(1)	251.029
Rh+3_H+1_NH3(5)_N3-1				
3	---	1	H(16)N(8)Rh(1)	231.091
Rh+3_H+1_PAR-2				
2	---	1	C(11)H(8)N(3)O(2)Rh(1)	317.113
Rh+3_H+1_TAR-2				
2	---	1	C(9)H(6)N(3)O(2)Rh(1)S(1)	323.136
Rh+3_H+1(2)_PAR-2(2)				
1	---	1	C(22)H(16)N(6)O(4)Rh(1)	531.317
Rh+3_Ile-1				
2	---	1	C(6)H(12)N(1)O(2)Rh(1)	233.077
Rh+3_Ile-1(2)				
1	---	1	C(12)H(24)N(2)O(4)Rh(1)	363.244
Rh+3_Leu-1				
2	---	1	C(6)H(12)N(1)O(2)Rh(1)	233.077
Rh+3_Leu-1(2)				
1	---	1	C(12)H(24)N(2)O(4)Rh(1)	363.244

Rh+3_Met-1				
2	---	1	C (5) H (10) N (1) O (2) Rh (1) S (1)	251.110
Rh+3_Met-1 (2)				
1	---	1	C (10) H (20) N (2) O (4) Rh (1) S (2)	399.310
Rh+3_NH3 (5)				
3	---	4	H (15) N (5) Rh (1)	188.063
Rh+3_NH3 (5)_Br-1				
2	---	1	Br (1) H (15) N (5) Rh (1)	267.967
Rh+3_NH3 (5)_Cl-1				
2	---	1	Cl (1) H (15) N (5) Rh (1)	223.516
Rh+3_NH3 (5)_N3-1				
2	---	1	H (15) N (8) Rh (1)	230.083
Rh+3_NH3 (5)_OH-1				
2	---	1	H (16) N (5) O (1) Rh (1)	205.070
Rh+3_NH3 (5)_SCN-1				
2	---	1	C (1) H (15) N (6) Rh (1) S (1)	246.140
Rh+3_NH3 (5)_SO4-2				
1	---	2	H (15) N (5) O (4) Rh (1) S (1)	284.120
Rh+3_OH-1				
2	---	6	H (1) O (1) Rh (1)	119.917
Rh+3_OH-1 (2)				
1	---	1	H (2) O (2) Rh (1)	136.925
Rh+3_OH-1 (3)_ (s) Rhodium(III) hydroxide				
0	21656-02-0	6	H (3) O (3) Rh (1)	153.932

Rh+3_OH-1 (4)				
-1	---	1	H (4) O (4) Rh (1)	170.939
Rh+3_Ser-1				
2	---	1	C (3) H (6) N (1) O (3) Rh (1)	206.996
Rh+3_Ser-1 (2)				
1	---	1	C (6) H (12) N (2) O (6) Rh (1)	311.081
Rh+3_Thr-1				
2	---	1	C (4) H (8) N (1) O (3) Rh (1)	221.022
Rh+3_Thr-1 (2)				
1	---	1	C (8) H (16) N (2) O (6) Rh (1)	339.135
Rh+3_Val-1				
2	---	1	C (5) H (10) N (1) O (2) Rh (1)	219.050
Rh+3_Val-1 (2)				
1	---	1	C (10) H (20) N (2) O (4) Rh (1)	335.190
Rh+4 Rhodium(IV) ion				
4	22537-45-7	2	Rh (1)	102.910
Rh+4_Cl-1 (6)				
-2	---	1	Cl (6) Rh (1)	315.628
Rh+6 Rhodium(VI) ion				
6	59596-47-3	1	Rh (1)	102.910
Rh (s) Rhodium				
0	7440-16-6	19	Rh (1)	102.910
Rh2O (s)				

0	---	1	O (1) Rh (2)	221.819
Rh₂O₃ (s) Rhodium oxide				
0	12036-35-0	4	O (3) Rh (2)	253.818
Rh₂S₃ (s) Rhodium Sulfide 2:3				
0	---	1	Rh (2) S (3)	302.000
Rh₃S₄ (s) Rhodium Sulfide 3:4				
0	---	1	Rh (3) S (4)	436.970
RhO (s)				
0	---	1	O (1) Rh (1)	118.909
RhO₂ (s)				
0	---	2	O (2) Rh (1)	134.909
RhO₄-2				
-2	---	1	O (4) Rh (1)	166.908
RhS_{0.882} (s) Rhodium(II) sulfide				
0	---	1	Rh (1) S (0.882)	131.187
S (s) Sulphur, alpha; Sulphur, orthorhombic; Sulphur; Sulfur, alpha; Sulfur, orthorhombic; Sulfur; Sulfur-Rhmb				
0	7704-34-9	619	S (1)	32.0600
SCN-1 Thiocyanate ion; NCS-1 equivalent				
-1	302-04-5	785	C (1) N (1) S (1)	58.0777
Ser-1 Serinate ion; L-Serinate ion; 2-amino-3-hydroxypropionate ion; beta-hydroxyalanate ion; 2-amino-3-hydroxypropanoate ion; alpha-amino-beta-hydroxypropionate ion; L-2-amino-3-hydroxypropanoate ion				
-1	17807-54-4	708	C (3) H (6) N (1) O (3)	104.086

SO4-2

Sulfate ion; Sulphate ion

-2	14808-79-8	1270	O(4)S(1)	96.0576
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Thr-1

Threoninate ion; L-Threoninate ion; 2-amino-3-hydroxybutyrate ion; alpha-amino-beta-hydroxybutyrate ion; 2-amino-3-hydroxybutanoate ion; L-2-amino-3-hydroxybutanoate ion

-1	68199-29-1	602	C(4)H(8)N(1)O(3)	118.112
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Val-1

L-Valinate ion; Valinate ion; L-2-Amino-3-methylbutanoate ion

-1	---	582	C(5)H(10)N(1)O(2)	116.140
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